Rajalakshmi Engineering College, Thandalam - 602105

(An Autonomous Institution, Affiliated to Anna University, Chennai)

Department of CSE

Zeroth Review PPT

T075 - An AI-Powered Voice Receptionist Framework for Smart Campus Communication and Information Access

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Project Domain:
Artificial Intelligence
Natural Language Processing

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Abstract

- Colleges receive large volumes of telephone inquiries from prospective students, parents, alumni & existing students
- Queries span admissions, academic programs, hostel facilities, fees, placements, examinations & student services
- Manual reception systems repeatedly answer the same questions, increasing workload & giving inconsistent responses
- Long waiting times & limited support outside working hours affect service quality
- High call volumes during admission & examination periods make communication management especially challenging

Abstract

- **Voice-Based AI Receptionist Agent** using RAG, LLMs, Speech-to-Text, Text-to-Speech & Agentic AI
- Retrieves verified information from a dedicated college knowledge base
- Supports natural, human-like voice conversations with callers
- Integrates with the college ERP after authentication for personalized services
- Intelligently transfers complex queries to a human receptionist when required
- Improves communication efficiency & delivers intelligent 24×7 assistance

Problem Statement

- **High call volumes, no time to answer it all.** Colleges get thousands of repetitive calls about admissions, academics, hostel, fees, placements and exams, and small reception teams can't keep up – especially during peak admission and exam periods.
- **Manual reception is slow and inconsistent.** Answers depend on whoever picks up the phone, so waiting times are long and the information given to callers isn't always accurate or consistent.
- **No support outside office hours.** Students and parents calling in the evening, on weekends, or during holidays get no response at all, leaving time-sensitive queries unresolved.

Solution

- **Develop a Voice-Based AI Receptionist** using **Speech-to-Text (STT), Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Text-to-Speech (TTS), and Agentic AI**.
- **Automatically answers routine queries** such as admissions, fees, departments, hostel, placements, events, and academic regulations, reducing receptionist workload. **Provides 24/7 support**, ensuring instant responses even outside office hours.
- **Delivers accurate and consistent information** by retrieving answers only from the official college knowledge base using RAG. **Supports natural voice conversations** by converting speech to text, generating context-aware responses, and replying through natural speech.
- Securely integrates with **DigiCampus** to provide personalized information (attendance, marks, fee status, exam results, and placement eligibility)
- **Intelligently escalates calls** to a human receptionist when confidence is low, the query requires administrative approval, or the caller requests human assistance.

Prototype Demonstration

Scan to Test the Voice AI Receptionist Prototype

PROTOTYPE KNOWLEDGE BASE

- Rajalakshmi Engineering College – an autonomous institution in Chennai, offering UG & PG engineering programmes

Departments

- CSE · IT · ECE · EEE · Mechanical · Civil · AI & DS · MBA

Facilities

- Hostel (Boys & Girls) · Library · College Transport · Training & Placement Cell

Authentication

- Attendance, Marks, Fees & Exam Results require ERP authentication before access

Current Prototype: Voice-based AI Receptionist built using ElevenLabs with a custom Knowledge Base
Future Enhancements: RAG + DigiCampus ERP + Twilio Calling + Human Call Transfer